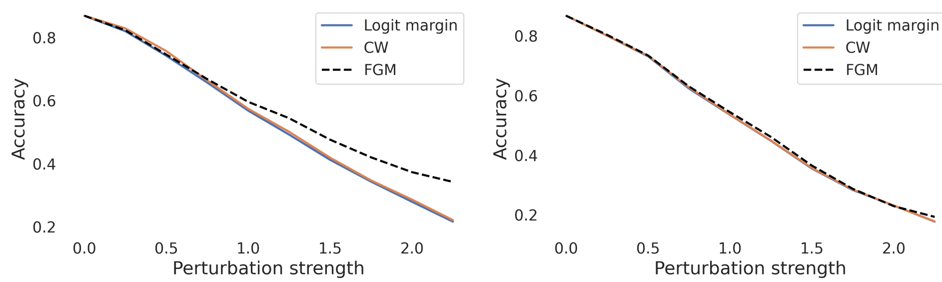
loss we used the Carlini-Wagner (CW) loss [Carlini and Wagner, 2017a] on the averaged logits, given by:

$$CW(x, \Theta^{\mathcal{A}}) = \max\left(\max_{i \neq t}(Z(x, \Theta^{\mathcal{A}})_i) - Z(x, \Theta^{\mathcal{A}})_t, 0\right) \quad ,$$

where $Z(x, \Theta^{\mathcal{A}})_t = \frac{1}{S^A} \sum_{s=1}^{S^A} Z(x, \theta_s)_t$ is an arbitrary averaged logit of an output node for input x. In figure 14 it is shown, that the attack with the CW loss on the infinite mixture model improves upon the original attack scheme (FGM), whereas it did not improve the attack's success for the BNN. We additionally conducted an attack based on the logit margin loss $\mathcal{L}(x + \delta^{\mathcal{A}}, y) = -(\min_{c \neq y} f_{y-c}(x))$ equivalent to the attack conducted on the SIN, but we found that it performs similar to the CW loss.

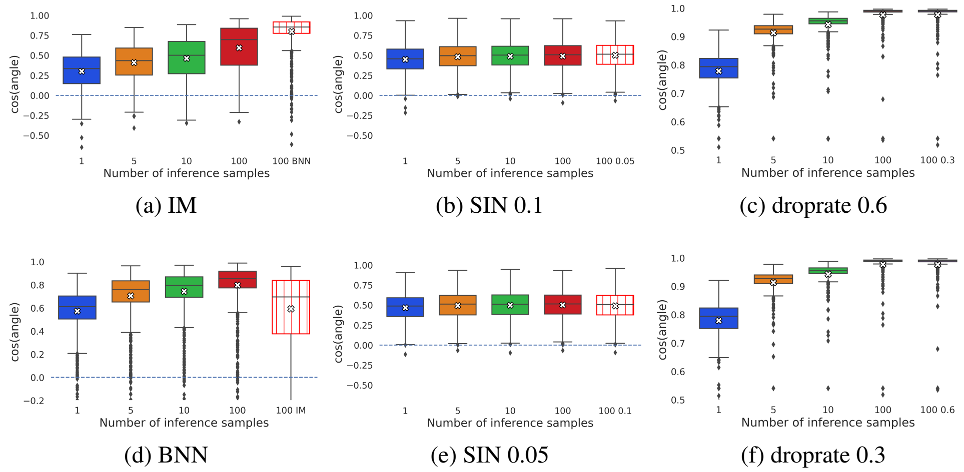


(a) IM (b) BNN

Figure 14: Accuracy under attack for the first 1,000 test set images from FashionMNIST with varying perturbation strengths and different attack objectives. Each attack was calculated based on 100 samples and ℓ_2-constraint for models trained on FashionMNIST.

C.4 Complementary experiments on robustness in dependence of the amount of samples used during inference

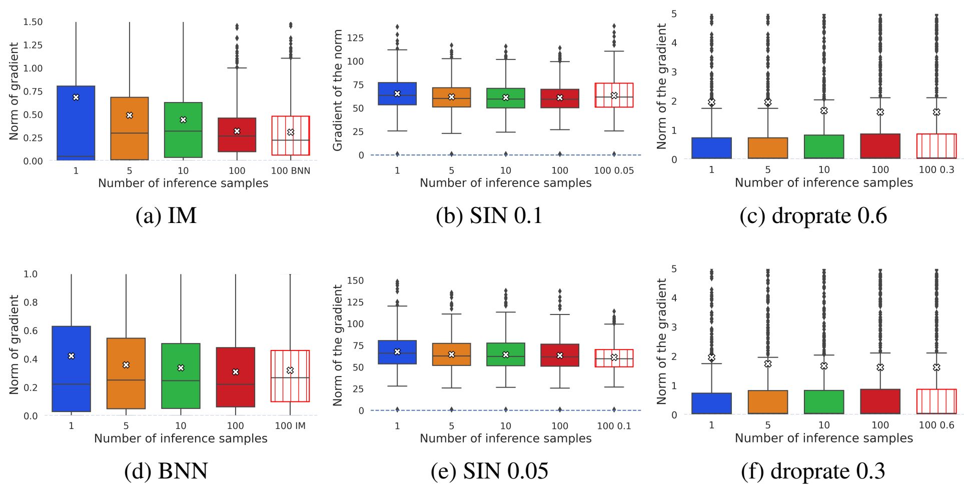
In the main part of the paper we argued why, surprisingly, the amount of samples during inference does not influence the robustness, even though we see in figure 15 that less sample lead to the smallest values for $\cos(\alpha_c^{\mathcal{I},\mathcal{A}})$. As stated in the main part, the increased gradient norm when using only few samples seems to compensate the assumed benefits with regard to the angle for using few samples (c.f. figure 16). This can also be seen in table 2 and figure 6 from the main paper, where a (negative) effect on the robustness can only be observed for one inference sample, which also leads to the worst test set accuracy. The benign prediction margins are also hardly effected by the increased number of samples during prediction (c.f. figure 17).



(a) IM (b) SIN 0.1 (c) droprate 0.6

(d) BNN (e) SIN 0.05 (f) droprate 0.3

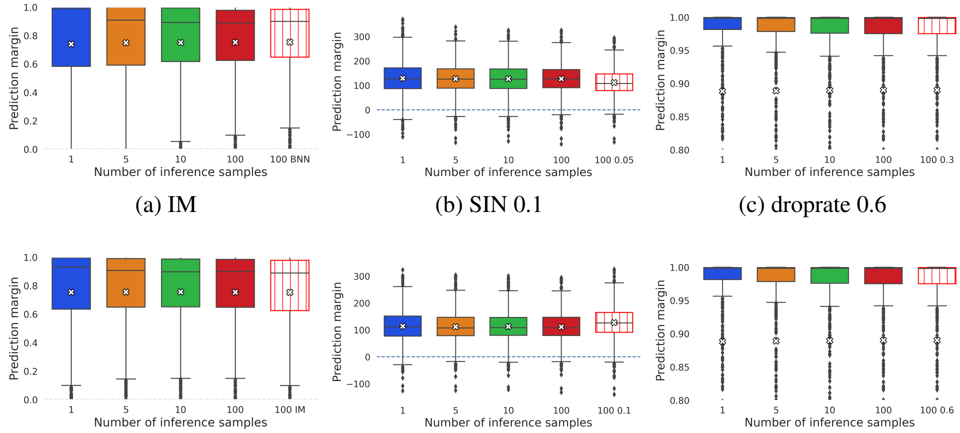
Figure 15: Cosine of the angle for models trained on FashionMNIST (a), b), d), e)) and trained on CIFAR10 (c) and f)) for different amounts of samples used during inference. Used attack direction δ was calculated based on 100 sample of FGM under ℓ_2-norm constraint with $\eta = 1.5$ and 0.3 respectively.



(a) IM (b) SIN 0.1 (c) droprate 0.6

(d) BNN (e) SIN 0.05 (f) droprate 0.3

Figure 16: Norm of the gradient for models trained on FashionMNIST (a), b), d), e)) and trained on CIFAR10 (c) and f)) for different amounts of samples used during inference.



(a) IM (b) SIN 0.1 (c) droprate 0.6

(d) BNN (e) SIN 0.05 (f) droprate 0.3

Figure 17: Prediction margin for the benign inputs for models trained on FashionMNIST (a), b), d), e)) and trained on CIFAR10 (c) and f)) for different amounts of samples used during inference.

C.5 Experiments for CIFAR100

For the experiments on CIFAR100 we used yet another method to create stochastic neural networks, namely by applying a Laplace approximation [MacKay, 1992] to an already trained network. This is archived by adapting a Gaussian distribution over the network's parameters such that the mean is given by the maximum a posterior estimate and the covariance are calculated to match the local loss curvature. Specifically, we used the GitHub library from Daxberger et al. [2021] on top of the adversarial trained wide ResNet70-16 with clean accuracy 69.15 provided by Gowal et al. [2020]. Because of the high amount of parameters in this network we used a last-layer diagonal Gaussian approximation for fitting our posterior distribution from which we sample the θ_i's for deriving an approximate expected prediction. As in the previous experiments we observe, that the adversarial accuracy decreases with more samples during attack, while the angle between the attack and negative gradient during inference decreases which leads to a cosine increase.

Table 3: Adversarial accuracy decrease and $\cos(\alpha_c^{\mathcal{I},\mathcal{A}})$ increase on CIFAR100 with increased amount of samples during attack, where the attack was conducted with ℓ_∞ norm constraint and perturbation strength 8/225.

# SAMPLES	ADVERSARIAL ACCURACY	AVERAGE $\cos(\alpha_c^{\mathcal{I},\mathcal{A}})\pm$ STD
1	48.00	0.2028 ± 0.112
5	45.00	0.2550 ± 0.100
10	43.90	0.2680 ± 0.095